

Computability: Exercise 6

Due: May 18th, 2026

Question 1

- Show that all non-trivial languages ($L \neq \emptyset$ and $L \neq \Sigma^*$) are R-hard, and find what is the set of R-complete languages.
- Let $\mathcal{C} \in \{\text{R}, \text{RE}, \text{coRE}\}$. Show formally that if L is a $\mathcal{C}$ -hard language and $L \leq_m K$ for some language K , then K is also $\mathcal{C}$ -hard.

Question 2

Show that $\text{REG}_{\text{TM}} \notin \text{RE}$.

Question 3

- Let $\text{ACC}_\epsilon = \{\langle M \rangle : M \text{ accepts when run on the input } \epsilon\}$. Show that ACC_ϵ is RE-complete.
- Show that $\text{SELF-NO-ACC}_{\text{TM}} \stackrel{\text{def}}{=} \{\langle M \rangle : M \text{ does not accept } \langle M \rangle\}$ is coRE-complete.
Hint: It may be useful to remember that a TM $\mathbf{r}$ is a reduction from L to L' if and only if it is a reduction from $\overline{L}$ to $\overline{L'}$.

Question 4

Classify each of the following languages into one of the classes R, $\text{RE} \setminus \text{R}$, $\text{coRE} \setminus \text{R}$, or $\overline{\text{RE} \cup \text{coRE}}$:

- $L \stackrel{\text{def}}{=} \{\langle M_1, M_2 \rangle : L(M_1) \cap L(M_2) = \emptyset\}$
- $L \stackrel{\text{def}}{=} \{\langle M_1, M_2 \rangle : \text{There exists } w \in \Sigma^* \text{ that } M_1 \text{ accepts and } M_2 \text{ rejects}\}$
- $L \stackrel{\text{def}}{=} \{\langle M \rangle : M \text{ rejects at least 10 different words}\}$ (recall that by “rejects” we mean that the machine actually reaches a rejecting final state)
- $L \stackrel{\text{def}}{=} \{(\langle M \rangle, 1^k) : M \text{ reaches at most } k \text{ different cells during its run on } \epsilon\}$

Question 5

Consider the language

$$L \stackrel{\text{def}}{=} \{\langle M, N, w \rangle : \langle M, w \rangle \in \text{HALT}_{\text{TM}}, \langle N, w \rangle \notin \text{HALT}_{\text{TM}}\},$$

and let $\mathcal{C} \stackrel{\text{def}}{=} \text{RE} \cup \text{coRE}$

- Show that L is $\mathcal{C}$ -hard with respect to $\mathcal{C}$.

b. Show that the class $\mathcal{C}$ is not closed under union.

Hint: Consider tiny modifications to the languages HALT_{TM} and $\text{NON-HALT}_{\text{TM}}$, and show that their union is $\mathcal{C}$ -hard.

c. Show that there is no complete language in $\mathcal{C}$.

Question 6

In class, we showed that the Halting Problem over every alphabet belongs to $\text{RE} \setminus \text{R}$.

Define HALT_{bin} as the Halting Problem over the binary alphabet, where $\Gamma = \{0, 1, \sqcup\}$.

Show that:

a. $\text{HALT}_{\text{TM}} \leq_m \text{HALT}_{\text{bin}}$

b. $\text{HALT}_{\epsilon} \leq_m \text{HALT}_{\epsilon, \text{bin}}$